

Technical Document #387: Natural Language Processing - Comprehensive Overview

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Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names.
- Machine translation converts text from one language to another.
- Word embeddings represent words as dense vectors in continuous space.